Pizza Sales Performance Analysis

SQL Data Analysis & Power BI Dashboard Project

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Project Overview

This project analyzes a full year of pizza sales transactions to uncover revenue drivers, customer ordering patterns, and product performance. The workflow combines SQL Server for data querying and KPI calculation with an interactive Power BI dashboard for visualization and business storytelling. The end goal is to translate raw transactional data into clear, actionable insights that support decisions on menu strategy, staffing, and promotions.

Business Questions Answered

- What are the overall revenue, order volume, and average order value?
- Which pizza categories and sizes generate the most sales?
- What are the best-selling and worst-selling pizzas by quantity?
- When are the busiest days and hours for orders?
- How do sales percentages break down by category and size over time?

SQL Analysis

Using SQL Server, the raw **pizza_sales** table was queried and aggregated to build four core analysis layers:

- **KPIs Query:** Total Revenue, Average Order Value, Total Pizzas Sold, Total Orders, and Average Pizzas per Order, calculated with SUM, COUNT(DISTINCT), and CAST for precise decimal formatting.
- **Trends Query:** Daily order trends using DATENAME(DW, order_date) and hourly order trends using DATEPART(HOUR, order_time), both ranked to surface peak ordering windows.
- **Percentages Query:** Contribution of each pizza category and size to total revenue, using correlated subqueries with MONTH() and DATEPART(quarter) filters, cast to decimal and formatted as percentage strings.
- **Top/Worst Sellers Query:** Top 5 and bottom 5 pizzas ranked by quantity sold using SELECT TOP with SUM and GROUP BY pizza_name.

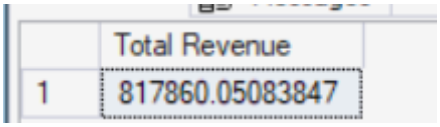
SQL Queries

-KPIs

- Total Revenue

```
-- Total Revenue
Select SUM (total_price) As 'Total Revenue'
from pizza_sales;
```

  - Result

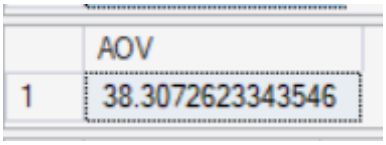


	Total Revenue
1	817860.05083847

- AOV (Average Order Value)

```
-- Total Revenue
Select SUM (total_price) As 'Total Revenue'
from pizza_sales;
```

  - Result

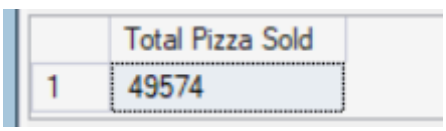


	AOV
1	38.3072623343546

- AOV (Average Order Value)

```
-- Total Pizza Sold
Select SUM (quantity) AS 'Total Pizza Sold'
from pizza_sales;
```

  - Result

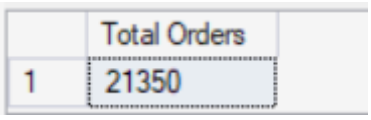


	Total Pizza Sold
1	49574

- Total Orders

```
-- Total Orders
Select COUNT (Distinct order_id) AS 'Total Orders'
from pizza_sales;
```

  - Result

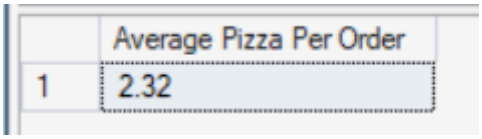


	Total Orders
1	21350

- Average Pizza Per Order

```
-- Average QTY Per Order
Select Cast(CAST(SUM (quantity) AS decimal(10,2))  /
CAST( COUNT (Distinct order_id)AS decimal(10,2))AS decimal(10,2)) AS 'Average Pizza Per Order'
from pizza_sales;
```

  - Result



	Average Pizza Per Order
1	2.32

-Trends

- Daily Trend

```
--Daily Trend
Select DATENAME (DW,order_date) as Order_Day , COUNT (distinct order_id) as 'Number Of Orders'
from pizza_sales
Group by DATENAME (DW,order_date)
Order by COUNT (distinct order_id) Desc;
```

  - Result

	Order_Day	Number Of Orders
1	Friday	3538
2	Thursday	3239
3	Saturday	3158
4	Wednesday	3024
5	Tuesday	2973
6	Monday	2794
7	Sunday	2624

- Hourly Trend

```
--Hourly Trend
Select DATEPART (HOUR,order_time) as Order_Hour , COUNT (distinct order_id) as 'Number Of Orders'
from pizza_sales
Group by DATEPART (HOUR,order_time)
Order by DATEPART (HOUR,order_time) ASC;
```

  - Result

	Order_Hour	Number Of Orders
1	9	1
2	10	8
3	11	1231
4	12	2520
5	13	2455
6	14	1472
7	15	1468
8	16	1920
9	17	2336
10	18	2399
11	19	2009
12	20	1642
13	21	1198
14	22	663
15	23	28

-Percentages

- Percentage Of Sales Per Category

```
--Percentage Of Sales Per Category

Select pizza_category , SUM(total_price) * 100 /
(select SUM(total_price) from pizza_sales Where MONTH (order_date) = 1) AS 'Percentage Of Sales %',
CAST(SUM(total_price)As decimal(10,2)) AS 'Total Revenue Per Category'
from pizza_sales
Where MONTH (order_date) = 1
group by pizza_category;
```

  - Result

	pizza_category	Percentage Of Sales %	Total Revenue Per Category
1	Classic	26.6779189176038	18619.40
2	Chicken	23.1952780348435	16188.75
3	Veggie	24.4370162489706	17055.40
4	Supreme	25.6897867985821	17929.75

- Percentage Of Sales By Pizza Size

```
--Percentage Of Sales By Pizza Size
Select pizza_size , CONCAT( CAST( SUM(total_price)*100 /
(select SUM(total_price) from pizza_sales Where DATEPART (quarter , order_date) = 1)  AS decimal (10,2) )  ,'%')
AS Percentage_Of_Sales_By_Pizza_Size ,
Cast(SUM(total_price) AS decimal (10,2)) AS 'Total Sales By Size'
From pizza_sales
Where DATEPART (quarter , order_date) = 1
group by pizza_size
Order By Percentage_Of_Sales_By_Pizza_Size DESC;
```

  - Result

	pizza_size	Percentage_Of_Sales_By_Pizza_Size	Total Sales By Size
1	L	46.37%	95229.65
2	M	29.78%	61159.00
3	S	22.10%	45384.25
4	XL	1.60%	3289.50
5	XXL	0.14%	287.60

-Performance

- Total Pizzas Sold By Category

```
-- Total Pizzas Sold By Category
Select pizza_category , Sum(quantity) As Num_Of_Pizzas_Sold
From pizza_sales
group by pizza_category
Order By Num_Of_Pizzas_Sold Desc;
```

- Result

	pizza_category	Num_Of_Pizzas_Sold
1	Classic	14888
2	Supreme	11987
3	Veggie	11649
4	Chicken	11050

- Top 5 Pizza Sold by Quantity

```
--Top 5 Pizza Sold by Quantity
Select Top 5 pizza_name , SUM(quantity) AS Num_Of_Pizzas_Sold
From pizza_sales
group by pizza_name
Order By Num_Of_Pizzas_Sold Desc;
```

- Result

	pizza_name	Num_Of_Pizzas_Sold
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

- Worst 5 Pizza Sold by Quantity

```
--Worst 5 Pizza Sold by Quantity
Select Top 5 pizza_name , SUM(quantity) AS Num_Of_Pizzas_Sold
From pizza_sales
group by pizza_name
Order By Num_Of_Pizzas_Sold ASC;
```

- Result

	pizza_name	Num_Of_Pizzas_Sold
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme ...	950
5	The Soppressata Pizza	961

Power BI Dashboard

The SQL outputs were brought into Power BI to build a single-page executive dashboard combining KPI cards, trend charts, category/size breakdowns, and a narrative panel of automated insights.



Pizza Sales Performance Dashboard - Power BI

Key Insights

Metric	Result
Total Revenue	$817.86K
Total Orders	21,350
Total Pizzas Sold	49,574
Average Order Value	$38.31
Avg. Pizzas per Order	2.32
Busiest Days	Friday & Saturday
Peak Hours	12PM-1PM and 5PM-8PM
Top Category	Classic (26.91% of sales)
Best Sellers	Classic Deluxe & Chicken BBQ
Worst Seller	The Brie Carre Pizza

Skills Demonstrated

SQL Server (aggregation, subqueries, date functions, ranking) | Data Cleaning & Transformation | Power BI (DAX-driven visuals, dashboard design) | KPI Definition & Business Reporting | Data Storytelling

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